RAIL INFRASTRUCTURE

Solutions for rail industry processes

Increase capacity / Improve safety / Improve reliability / Reduce costs

Getting automation on track

The railway system in the UK is the oldest in the world, dating back to 1825. It now boasts more than 15,000 km of lines of which around a third are electrified. It is one of the busiest rail networks in the world.

The UK rail industry continues to undergo change, especially related to control and automation systems. It is moving away from restrictive engineering standards which added disproportionate costs when purchasing new equipment and has moved to adopt standard COTS (commercially off the shelf) products and services.

Outsourcing of maintenance and service contracts has led to demands on the automation control systems to provide remote access to diagnostic and performance information, easily accessible to managers. Increased information from assets has led to shortened downtime and increased operational efficiency analysis.

Challenges

Every asset controlled by a programmable device has the capacity of delivering a wealth of information about its performance and vital information for the efficient running of the enterprise. All assets contain "data" which may or may not be being utilised further up the chain.

In many cases this data is simply lying dormant in the local control device. However, retrieving this data is not as easy as it may appear to be.

Historically, disparate control systems existed with different network protocols and this made establishing common network architectures difficult, if not impossible.

Standards

Innovative automation techniques can overcome these obstacles but the key to unlocking this dormant information is the adoption of standards.

This can range from network protocols through programming standards and compliant secure databases.

Standards allow choice and flexibility and future proof data collection and control going forward.

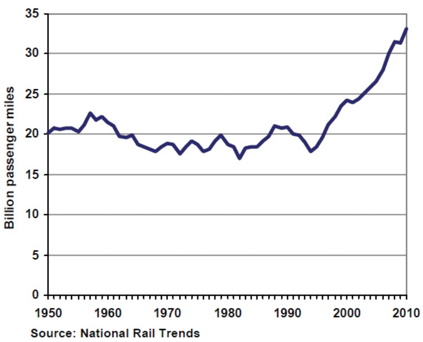
Innovation

Mitsubishi Electric delivers innovative leading edge technology which provides an opportunity to "rethink" traditional automation techniques and architectures and challenge the engineering practices of the past.

Mitsubishi in rail

Mitsubishi Electric has been involved in the UK Rail Industry since the late 1980s and has been and continues to be involved with mainline, underground and over ground projects and is a registered and approved supplier.

Investing in the future



Rail passengers miles: UK 1950 to 2012

Growth in Britain's railways

Britain has one of the fastest growing railway systems in Europe. In 2012/13 rail passengers made over 1.5 billion journeys travelling over 36 billion miles. This has doubled since the mid 1990's. On London Underground over 1.4 billion passengers were reported in 2012/13 covering over 46 million miles. Transferring road traffic to rail is a good thing for reducing congestion and carbon emissions however, a 2% shift from road to rail requires a 25% increase in rail capacity.

Investment

investment has been made and considerable infrastructure spend is planned for the future in order to maintain and improve the service and prevent overcrowding. During 2014-2019 (CP5) Network Rail is investing £21 billion. Transport for London plan an annual spend of £1.4 billion for the Underground and Crossrail with a further £4 billion planned investment up to 2016.

Key sector drivers

This unprecedented growth and planned investment is required to meet the key sector drivers of:

- Increasing capacity
  Longer, faster trains
  Electrification projects
  Moving block signalling concept
  European Compliant Train Control System (ECTS)
- Improving safety
  Automatic Train Protection Systems (ATP)
- Improving reliability and performance
  Remote Condition Monitoring (RCM)
  ORBIS concept (Offering Rail Better Information Services)
- Reducing operating costs
  Introducing Central Control & Communications Centres
  Reducing staff costs
- Sustainability
  Increased use of electrified trains
  Reduce carbon emissions
  Energy efficient assets

Achieving the digital railway

Collaboration and innovation between consortia in the rail sector ecosystem is seen as key in developing new data driven applications that help meet the Industry's key drivers and vision for the future of the Digital Railway.

The Technical Strategy Leadership Group's (TSLG) 30 year vision for rail strategy encapsulates this approach in defining innovative information and data systems that can be harnessed to deliver operation savings across the sector.

Railway ecosystem

Technology and innovation are seen as key enablers to meeting the drivers for the Rail Sector. Mitsubishi Electric plays its part in the supplier ecosystem providing leading edge solutions, based upon commercially off the shelf (COTS) products for telemetry, control, data management and visualisation that address the needs of the rail infrastructure sector.

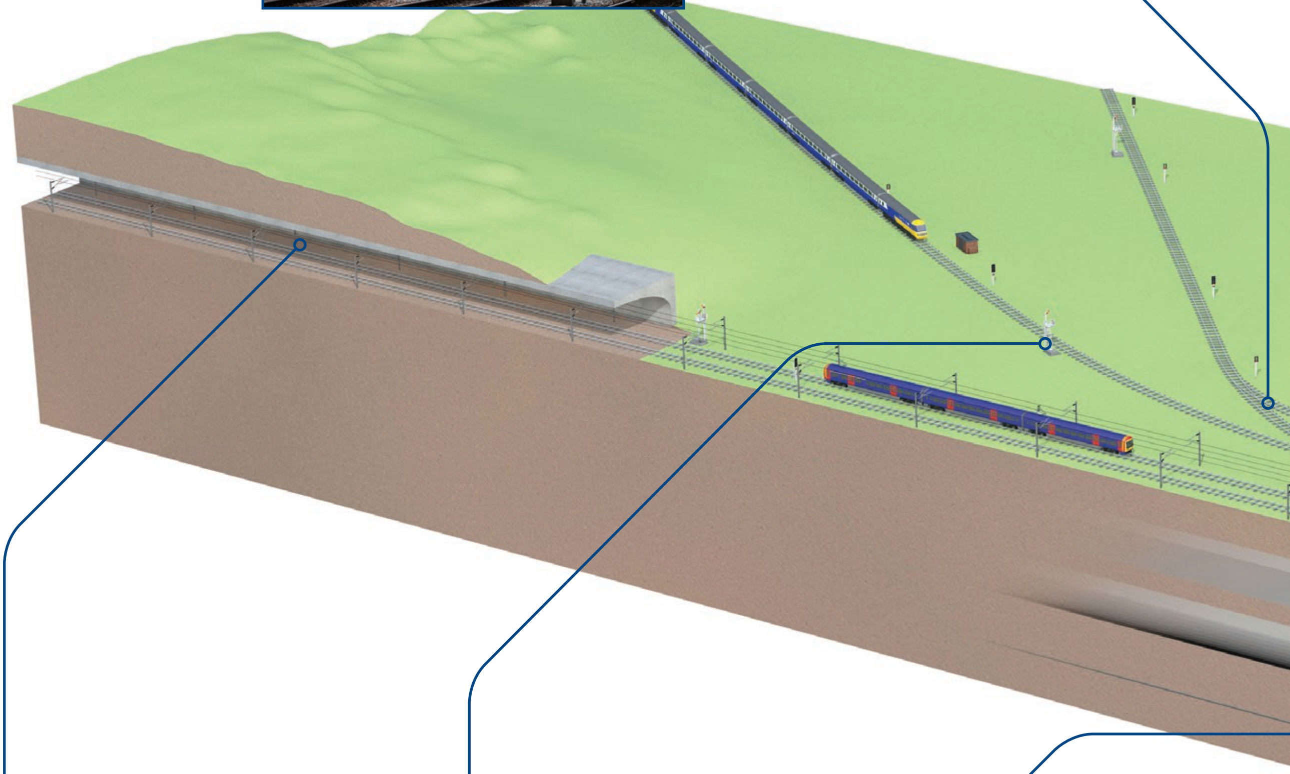
Mitsubishi Electric work in collaboration with many of the industry's leading companies in the UK including:

- Network Rail
- London Underground and Tubelines
- Thales Signalling Solutions
- Siemens Rail
- Vital Rail
- Firstco
- Hima Sella
- Johnson Control

Managing rail industry processes

Trackside control & monitoring

With a proven track record for reliability and security, Mitsubishi Electric compact and modular PLCs are used in a diverse range of trackside control & asset monitoring applications including the use of embedded secure data management technologies from industry leaders such as Raima Inc.



Emergency tunnel ventilation

Provision of Ventilation Control & Supervisory Systems using high availability redundant iQ modular PLCs, VSDs and MAPS SCADA.

Signalling

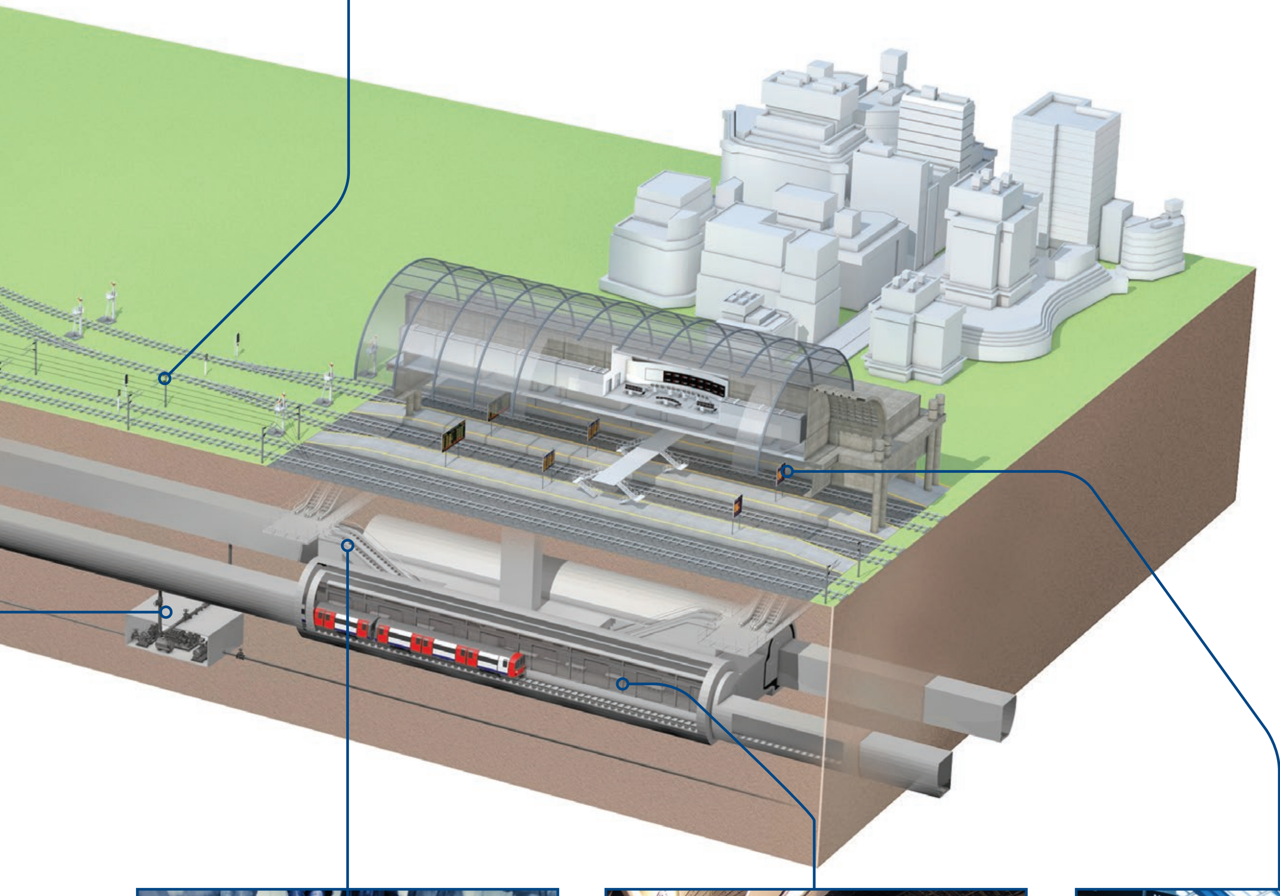
iQ Modular PLCs used to provide diversity in signalling control applications and compact PLCs used for power control of Principle Supply Points.

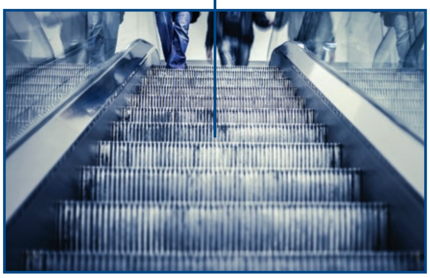
De-watering

Variable Speed Drives (VSDs) for Pump Control.

Electrification

iQ modular PLC, supporting industry standard open protocols such as IEC 61850, are used in traction substation automation including RTUs, bay control and complex logic for switching IED's.





Escalators & lifts

A complete portfolio of product offered for the control of lifts and escalators covering HMIs for local visualisation with modular PLCs and VSDs for control.

Platform edge doors

Mitsubishi Electric Compact FX PLC range used for door control logic.

Station communication & annunciation

iQ modular PLCs used for alarm management and Mitsubishi MAPS SCADA used for visualisation as part of the scope of station management, depot control & communications systems.

Electrification

Electrification

Electrification is a key process as part of the driver to increase capacity and sustainability of the UK rail network. At present some 40% of the existing network in the UK is electrified and government is committed to investing in programmes to increase this in coming control periods. Most of the electrified network is fed via overhead lines with the remainder being 3rd rail systems. The Department for Transport's Rail Technical Strategy states that "the right long term solution for rail would be one that minimises its carbon footprint and energy bill". Electrification, although having a high capital cost, is a low risk, mature and available technology, that is able to assist in achieving this.

Benefits of electrification: faster longer trains

Electric trains have more seats than the same length diesel trains and can pull more carriages. Electric trains typically have more than twice the traction power of diesel trains. Journey times are made shorter through increased speed of an electrified service.

A positive impact on the environment

The use of electric trains has a green impact on the environment with a reduction in carbon emission per passenger reduced by around 30% over diesel trains, thus improving air quality in our urban areas and around stations.

Reduction in operating costs

The use of electric trains helps to reduce operating costs in a number of ways, including the use of regenerative braking and improving the efficiency of electrified system, which are lighter and reduce damage to the rail infrastructure. Typically for passenger services, fuel cost savings are around 20% per passenger mile.

Solutions for Electrification

Mitsubishi Electric provides solutions covering an important cornerstone of the electrification process as part of the Railway Technical Strategy defined by the DfT, namely the optimisation of traction power and energy consumption through the management of electrical power distribution networks which includes control centre SCADA solutions for visualisation and control of the network and telemetry products for substation automation.

Control Centre SCADA Solutions

Mitsubishi MAPS SCADA solutions offer a range of benefits as part of an electrical switching control centre application including:

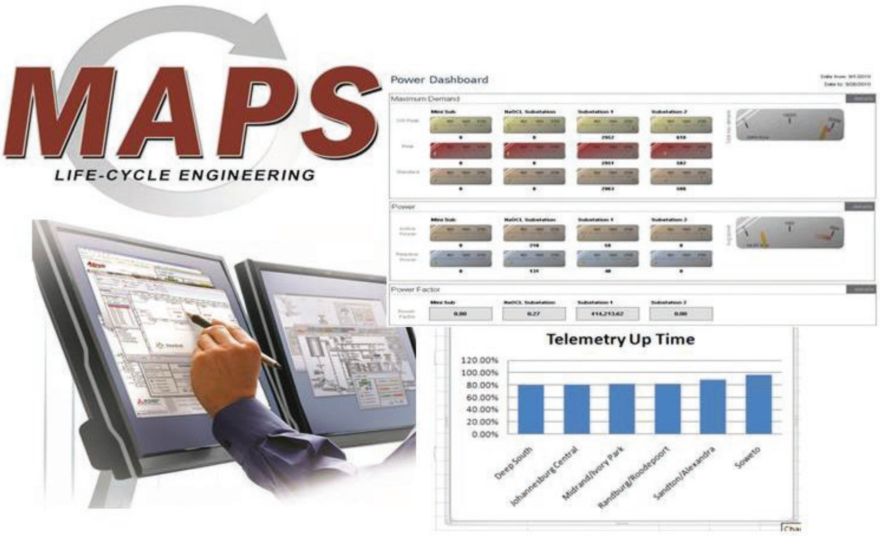
- Reduced development, engineering and lifecycle costs through the provision of an integrated development environment
- Production of lifecycle development documentation for achieving Cenelec standards such as EN 50126 and EN 50128
- Utilisation of typical industry communications standards such as DNP 3 and IEC 60870-5-104 for control centre/substation connectivity

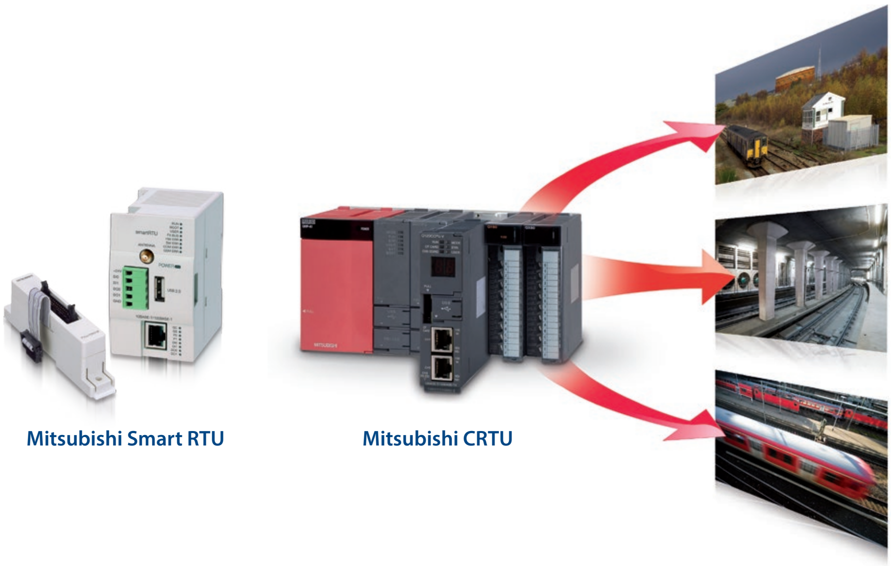
Telemetry Solutions

Mitsubishi Electric's telemetry solutions cover a range of PLC based RTU products from a cost effective offering providing DNP 3 communications up to a modular PLC based solution providing a range of open communications protocols prevalent in the industry including:

- DNP 3
- IEC 60870
- KEMA certified IEC 61850 (client 7 server) & GOOSE Messaging

The above, in conjunction with the use of Substation Control Language (SCL) and IEC 61131 PLC programming constructs, offer all the benefits of a PLC based RTU including being open, COTS (Commercially Off The Shelf) and the ability to be used in non-vendor specific applications.





Mitsubishi Smart RTU

Mitsubishi CRTU

Signalling

Demand for modern signalling

The re-signalling of the rail network is driven by the need to increase capacity in order to meet growing demand in the use of the railways and goes hand in hand with the driver to increase safety of the network. As part of the process to electrify the network, new signalling systems are required to cope with the longer and faster trains that are deployed on the electrified networks. Some of the other benefits of modern signalling solutions include:

- Reduced number of train delays
- Increase in speed of service
- Increase in number of trains operating a route
- Reduction in operations and maintenance costs
- Improvement in information for passengers

Signalling solutions

Mitsubishi Electric provides solutions covering several different aspects of the overall signalling system and this includes:

- Trackside data collection
- Monitoring of signal interlocking systems
- Control and telemetry for principal supply points

A vital component

Signalling is one of the most important components of a railway system. Train movement safety depends on it and the control and management of trains depends on them. Over the years many signalling and train control systems have evolved, with signalling systems developing from fixed block , track circuit based solutions to moving block , modern Communications Based Train Control Systems (CBTC) that offer trackside to/from train data communications that facilitates the development of function such as:

- Automatic Train Protection (ATC)
- Automatic Train Operation (ATO)
- Automatic Train Supervision (ATS)

as defined by IEE 1474. So, today, a highly technical and complex industry has developed surrounding the provision of rail signalling systems

Secure trackside data collection

The rail industry has struggled for years with the problem of integrating efficient data management within its applications. Mainly due to the strict safety and security requirements which are normally extremely costly to meet. Developers of these systems incur large costs in analysis and testing as well as automation software to prevent any possibility of system catastrophic failure.

As a long term, proven supplier, to both surface and sub-terrain rail transit, Mitsubishi Electric's PLC systems have proven themselves reliable and safe over many applications and with the iQ Platform, in conjunction with an ACID compliant integrated secure database Mitsubishi provide a solution that offers:

- Secure data repository in a high availability PLC based environment
- Secure data transactions and connectivity across disparate systems
- Removal of "PC based" systems
- System that is capable of inclusion in an IEC 61508, SIL 2 system

Signal interlocking

As a recognised supplier of reliable, safe and COTS (commercially off the shelf) based programmable logic controllers (PLCs) for rail infrastructure applications, Mitsubishi Electric's controllers have been used extensively on rail signal interlocking systems as an independent monitor of the sequential operation of track circuits, adding an additional level of system diversity and safety integrity to computer based rail signal interlocking applications (CBI).

Principal supply points

Of critical importance in implementing rail signalling systems is ensuring availability of the power supply at all times. This is typically achieved through the use of Principal Supply Points (PSP's) which switch between power connections supplied by the Distribution Network Operator (DNO) and back up emergency power generation.

Traditionally, standby generator sets used for signalling systems on the over ground rail infrastructure had used hard wired interlock systems with standard control gear. These have been replaced by the use of programmable controller technology.

Key benefits

The benefits of this are reduced maintenance, increased flexibility and the ability to collate alarm conditions and other information and send this back via remote connection technology, for analysis and action.

Mitsubishi Electric PLCs such as the FX range offer a low cost, reliable control philosophy for standby generator sets and can be combined with Mitsubishi Electric Telemetry products such as Smart RTU and C RTU to provide wide area communications with control centres.

Point to point solutions

Mitsubishi provide solutions for point to point control of trackside assets such as:

- MOS (Motorised Switches)
- FED (Fixed Earthing Devices)
- CTS (Controllable Track Switches)

Based on COTS products and open industry standard communications networks and protocols such as CC-Link and Modbus.

Asset monitoring

Opportunities

Every asset controlled by a programmable device has the capacity to deliver a wealth of information about its performance and vital information, for example, for predictive maintenance techniques and other system critical operations.

Untapped resource

All assets contain data which may or may not be being utilised further up the chain. In many cases this data is simply lying dormant in local control devices but represents a "gold mine" of information about the state of asset operations. A flexible and innovative approach can unlock this data and change it into vital information that serves the needs of the organisation.

Infrastructure

An obstacle to gathering data can be that network architectures do not always exist and there are disparate systems to gather data from. Flexible control platforms and adoption of standard network communication protocols can help to overcome this. Mitsubishi Electric provide interfaces to Rail Industry standard protocols such as IEC 61850 and DNP3 as well as many other standard protocols which makes interfacing to other vendor's equipment easy and cost effective.

Data management

Mitsubishi Electric provides innovative, leading edge, commercially off the shelf (COTS) technology which provides the ability to interface directly from the asset to business level systems. Having unlocked the dormant asset data this information can start to deliver real value to the organisation from operational efficiency improvements through to energy saving and predictive asset maintenance. A greater knowledge of what is happening at the asset level will help to meet the organisation's business drivers.

Cyber security

Challenges

Cyber terrorism has become ever more sophisticated and is a real threat to the National Infrastructure. Infiltrating an organisation is a multi–layered approach, with bad practice and disgruntled personnel posing as big a threat as "traditional" attacks from the internet.

Change in the landscape

Often the "top end" of any organisation is well protected by sophisticated firewall and other protection systems. However with the advent of malware such as stuxnet, the point of attack switched to the generally more vulnerable automation layer and exposed weaknesses in some automation architectures.

Convergence

Convergence and adoption of new technologies such as IP based systems and architectures that include gateway PCs to link between the asset level and business systems has provided opportunities to cyber terrorists but these threats can be mitigated against.

Innovation going forward

A "rethink" of traditional architectures can help to reduce the security risk and probability of attack. At Mitsubishi Electric we deliver innovative leading edge technology which challenges the traditional system topology and provides direct, non PC based connection from the asset to the business level.

Safety

Passenger movement

Mitsubishi Electric has been heavily involved in the London Underground Escalator & Lift refurbishment programme. This programme has delivered leading edge technology, increased reliability, safety and cost savings

Ventilation and air management

This is clearly a high priority from a control and safety perspective. Mitsubishi technology has been used in innovative methods of air and pressure management in areas of the Underground network and also in the Channel Tunnel Rail Link (CTRL) tunnels. Air flow management systems diverted harmful fumes away from public areas and escape routes.

SIL rated standard control systems

Demand has grown for project control systems to meet IEC 61508 and associated Safety Integrity Levels (SIL). SIL levels apply to the whole control system and Mitsubishi Electric is able to supply Mean Time Between Failures(MTBF) and Failure Rate data for our equipment for inclusion in the system safety case. Mitsubishi equipment MTBF data is calculated using the MIL-HDBK-217 military standard.

Quality and reliability are the cornerstones of Mitsubishi product development and our standard control product solutions have been used in systems that have achieved up to SIL level 2 within the Rail Industry.

Safety Systems

Mitsubishi also have dedicated safety solutions, ranging from an integrated safety relay, through a mid-range programmable relay system, to a full blown safety PLC. These systems meet ISO 13849-1 PLe and the safety PLC is certified for use in systems pertaining to IEC 61508 SIL 3.

EMC Standards

Mitsubishi Electric solutions conform to railway EMC standards including EN50121 and London Underground E1027.

A world of automation solutions

Mitsubishi Electric offers a wide range of automation equipment from PLCs and HMIs to Robots and Inverters.

A name to trust

Since its beginnings in 1870, some 45 companies use the Mitsubishi Electric name, covering a spectrum of finance, commerce and industry.

The Mitsubishi Electric brand name is recognised around the world as a symbol of premium quality.

Mitsubishi Electric Corporation represents space development, transportation, semiconductors, energy systems, communications and information processing, audio visual equipment, home electronics, building and energy management and automation systems, it has 237 factories and laboratories worldwide in over 121 countries.

This is why you can rely on a Mitsubishi Electric automation solution – because we know first hand about the need for reliable, efficient, easy-to-use automation and control.

As one of the world's leading companies with a global turnover of 4 trillion Yen (approximately $40 billion), employing over 100,000 people, Mitsubishi Electric has the resource and the commitment to deliver the ultimate in service and support as well as the best products.

Service & Support

As with any activity, maintaining the quality and operational function of your equipment is essential. Downtime from any operational failure is never good news. In today's tough business conditions returning to full production as soon as possible is critical. At Mitsubishi Electric we aim to deliver industry leading levels of service and support that will provide cost savings, improved machine availability and system uptime to our customers and minimise implementation risk.

We provide a professional on-site service through our System Service Team and Service Agent network and offer a tailored range of service contracts for single and multisite companies, on an annual or project basis called our 3 Diamond Service.

- Dedicated technical support
- 24/7 on site engineer call-out availability
- Annual maintenance visits
- Extended warranty
- Technical manual library
- Multi-vendor product support
- Legacy product survey and risk assessment
- Commissioning service
- Repair services

Global partner. Local friend.

Version check

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